

2024 年春高等数学 B2 期中考试参考答案

一、1. C 2. B 3. D 4. D 5. A

二、6. $\sqrt{14}$ 7. $\frac{\pi}{6}$ 8. $y = Ce^{-x^2} + \frac{1}{2}$ 9. $\frac{5\pi}{4}$ 10. $\frac{1}{3}dx + \frac{1}{3}dy + \frac{1}{3}dz$

三、11. 令 $p = y'$, 得 $\frac{dp}{dx} = 1 + p^2$. (2 分)

$\arctan p = x + C_1$, 即 $y' = \tan(x + C_1)$, 由 $y'(0) = 0$ 得 $C_1 = 0$. (2 分)

$y' = \tan x$, $y = -\ln|\cos x| + C_2$, 由 $y(0) = 0$ 得 $C_2 = 0$, $y = -\ln|\cos x|$. (2 分)

12. 过点 $A(3,1,5)$ 与直线 $L: \frac{x}{1} = \frac{y}{2} = \frac{z+2}{-2}$ 垂直的平面方程为 $(x-3) + 2(y-1) - 2(z-5) = 0$, (2 分)

该平面方程与直线方程联立解得 $(x, y, z) = (-1, -2, 0)$, (2 分)

此为点 A 在直线上 L 上的投影点 (记为 B) 的坐标, 则所求距离为 $|\overline{AB}| = \sqrt{4^2 + 3^2 + 5^2} = 5\sqrt{2}$. (2 分)

13. $\frac{\partial z}{\partial u} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial u} = \frac{1}{1 + \left(\frac{x}{y}\right)^2} \frac{1}{y} + \frac{1}{1 + \left(\frac{x}{y}\right)^2} \left(-\frac{x}{y^2}\right)$, (2 分)

$\frac{\partial z}{\partial v} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial v} = \frac{1}{1 + \left(\frac{x}{y}\right)^2} \frac{1}{y} + \frac{1}{1 + \left(\frac{x}{y}\right)^2} \left(-\frac{x}{y^2}\right) (-1)$, (2 分)

$\frac{\partial z}{\partial u} + \frac{\partial z}{\partial v} = \frac{2y}{x^2 + y^2} = \frac{u-v}{u^2 + v^2}$. (2 分)

14. 对 x 求导, 得 $\begin{cases} 2x + 2y \frac{dy}{dx} + 2z \frac{dz}{dx} = 0 \\ 1 + \frac{dy}{dx} - \frac{dz}{dx} = 0 \end{cases}$, 点 $(1,1,1)$ 处, $\begin{cases} \frac{dy}{dx} + \frac{dz}{dx} = -1 \\ \frac{dy}{dx} - \frac{dz}{dx} = -1 \end{cases}$, 解得 $\begin{cases} \frac{dy}{dx} = -1 \\ \frac{dz}{dx} = 0 \end{cases}$, (3 分)

点 $(1,1,1)$ 处的切向量为 $\left(1, \frac{dy}{dx}, \frac{dz}{dx}\right) = (1, -1, 0)$, 点 $(1,1,1)$ 处的切线方程: $\frac{x-1}{1} = \frac{y-1}{-1} = \frac{z-1}{0}$,

法平面方程: $(x-1) - (y-1) = 0$, 即 $x - y = 0$. (3 分)

15. 将 $y = e^x$ 代入 $y'' + py' + 6y = 2e^x$ 得, $e^x + pe^x + 6e^x = 2e^x$, $p = -5$, (2 分)

由特征方程 $r^2 - 5r + 6 = 0$ 得 $r_1 = 2$, $r_2 = 3$, $y'' - 5y' + 6y = 0$ 的通解为 $C_1 e^{2x} + C_2 e^{3x}$, (2 分)

$y'' - 5y' + 6y = 2e^x$ 的通解为 $C_1 e^{2x} + C_2 e^{3x} + e^x$. (2 分)

16. 交换积分次序 $\int_0^1 dx \int_x^1 \sin y^2 dy = \int_0^1 dy \int_0^y \sin y^2 dx$ (2分)

$$= \int_0^1 y \sin y^2 dy = \frac{1}{2} \int_0^1 \sin y^2 dy^2 \quad (2分)$$

$$= -\frac{\cos y^2 \Big|_0^1}{2} = \frac{1 - \cos 1}{2}. \quad (2分)$$

17. L_1 的方向向量 $\mathbf{s}_1 = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 1 & 1 & 3 \\ -1 & 3 & 5 \end{vmatrix} = (-4, -8, 4) = -4(1, 2, -1),$

$$L_2 \text{ 的方向向量 } \mathbf{s}_2 = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -3 & 2 & 1 \\ 5 & 1 & 7 \end{vmatrix} = (13, 26, -13) = 13(1, 2, -1), \quad \mathbf{s}_1 // \mathbf{s}_2, \text{ 所以 } L_1 \text{ 与 } L_2 \text{ 平行. (3分)}$$

L_1 上任取一点, 不妨为 $A(2, 1, 0)$, L_2 上任取一点, 不妨为 $B(0, 4, 0)$, 所求平面的法向量为

$$\overrightarrow{AB} \times \mathbf{s}_1 = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -2 & 3 & 0 \\ 1 & 2 & -1 \end{vmatrix} = (-3, -2, -7) = -(3, 2, 7), \quad (3分)$$

所求平面的方程为 $3x + 2(y - 4) + 7z = 0$, 即 $3x + 2y + 7z - 8 = 0$. (1分)

18. 联立 $z = 2 - x^2 - y^2$ 与 $z = \sqrt{x^2 + y^2}$ 得 $x^2 + y^2 = 1$, (2分)

$$\text{所求体积为 } \iiint_{\Omega} dv = \int_0^{2\pi} d\theta \int_0^1 \rho d\rho \int_{\rho}^{2-\rho^2} dz \quad (3分)$$

$$= 2\pi \int_0^1 \rho(2 - \rho^2 - \rho) d\rho = \frac{5\pi}{6}. \quad (2分)$$

19. $f(x) = \sin x - x \int_0^x f(t) dt + \int_0^x t f(t) dt, \quad f'(x) = \cos x - \int_0^x f(t) dt, \quad f''(x) = -\sin x - f(x),$

$$f''(x) + f(x) = -\sin x, \text{ 特征方程为 } r^2 + 1 = 0, \quad r = \pm i,$$

$$f''(x) + f(x) = 0 \text{ 的通解为 } f(x) = C_1 \cos x + C_2 \sin x, \quad (3分)$$

$$\lambda + i\omega = i \text{ 是特征方程的根, } f''(x) + f(x) = -\sin x \text{ 的特解为 } y^* = x(a \cos x + b \sin x),$$

$$\text{代入方程 } f''(x) + f(x) = -\sin x \text{ 得 } a = \frac{1}{2}, \quad b = 0, \quad y^* = \frac{1}{2} x \cos x, \quad (2分)$$

$$f''(x) + f(x) = -\sin x \text{ 的通解为 } f(x) = C_1 \cos x + C_2 \sin x + \frac{1}{2} x \cos x,$$

$$\text{又 } f(0) = 0, \quad f'(0) = 1, \text{ 得 } C_1 = 0, \quad C_2 = \frac{1}{2}, \text{ 所以 } f(x) = \frac{1}{2}(\sin x + x \cos x). \quad (2分)$$

20. 设所求平面的方程为 $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$, 则 $\frac{1}{a} + \frac{2}{b} + \frac{3}{c} = 1$,

该平面与三个坐标面所围立体的体积为 $V = \frac{1}{6}abc$, (3 分)

$$\text{令 } F = abc + \lambda \left(\frac{1}{a} + \frac{2}{b} + \frac{3}{c} - 1 \right), \text{ 由 } \begin{cases} F_a = bc - \frac{\lambda}{a^2} = 0 \\ F_b = ac - \frac{2\lambda}{b^2} = 0 \\ F_c = ab - \frac{3\lambda}{c^2} = 0 \\ \frac{1}{a} + \frac{2}{b} + \frac{3}{c} = 1 \end{cases} \quad (3 \text{ 分})$$

得 $(a, b, c) = (3, 6, 9)$, 由问题的实际意义可知, 在该点处所讨论的体积取得最小值, 所求平面方程为

$$\frac{x}{3} + \frac{y}{6} + \frac{z}{9} = 1. \quad (2 \text{ 分})$$

21. 连续.

$$0 \leq \frac{x^2 y^2}{\sqrt{x^2 + y^2}} \leq \frac{\frac{1}{4}(x^2 + y^2)^2}{\sqrt{x^2 + y^2}} = \frac{1}{4}(x^2 + y^2)^{\frac{3}{2}} \rightarrow 0, \quad (x, y) \rightarrow (0, 0),$$

$$\text{得 } \lim_{(x, y) \rightarrow (0, 0)} \frac{x^2 y^2}{\sqrt{x^2 + y^2}} = 0 = f(0, 0). \quad (2 \text{ 分})$$

偏导数存在.

$$\lim_{x \rightarrow 0} \frac{f(x, 0) - f(0, 0)}{x} = \lim_{x \rightarrow 0} \frac{0 - 0}{x} = 0, \quad f_x(0, 0) = 0, \text{ 同理, } f_y(0, 0) = 0. \quad (2 \text{ 分})$$

可微.

$$\lim_{(\Delta x, \Delta y) \rightarrow (0, 0)} \frac{f(0 + \Delta x, 0 + \Delta y) - f(0, 0) - [f_x(0, 0)\Delta x + f_y(0, 0)\Delta y]}{\sqrt{(\Delta x)^2 + (\Delta y)^2}} = \lim_{(\Delta x, \Delta y) \rightarrow (0, 0)} \frac{(\Delta x)^2 (\Delta y)^2}{(\Delta x)^2 + (\Delta y)^2},$$

$$\text{而 } 0 \leq \frac{(\Delta x)^2 (\Delta y)^2}{(\Delta x)^2 + (\Delta y)^2} \leq \frac{\frac{1}{4}[(\Delta x)^2 + (\Delta y)^2]^2}{(\Delta x)^2 + (\Delta y)^2} = \frac{1}{4}[(\Delta x)^2 + (\Delta y)^2] \rightarrow 0, \quad (\Delta x, \Delta y) \rightarrow (0, 0),$$

$$\text{所以 } \lim_{(\Delta x, \Delta y) \rightarrow (0, 0)} \frac{(\Delta x)^2 (\Delta y)^2}{(\Delta x)^2 + (\Delta y)^2} = 0, \quad \Delta z - [f_x(0, 0)\Delta x + f_y(0, 0)\Delta y] = o(\rho). \quad (2 \text{ 分})$$